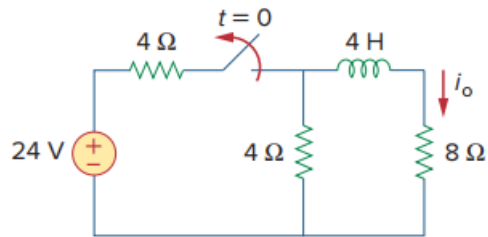


Problem

For the circuit in Fig. 7.91, find i_o for $t > 0$.

Fig. 7.91:



Step-by-step solution

Step 1 of 8

Consider the following circuit diagram:

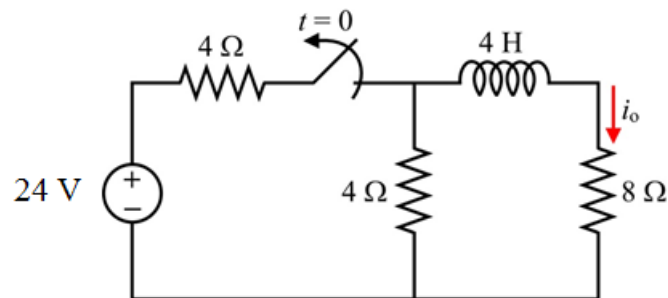


Figure 1

Step 2 of 8

For $t = 0^-$, the switch in the circuit is closed. Inductor acts as short circuit to dc. Draw the modified circuit diagram.

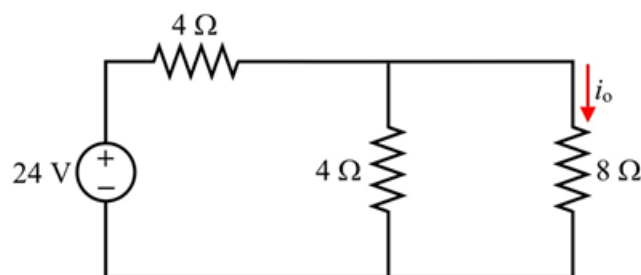


Figure 2

Step 3 of 8

Draw the above circuit with mesh notations.

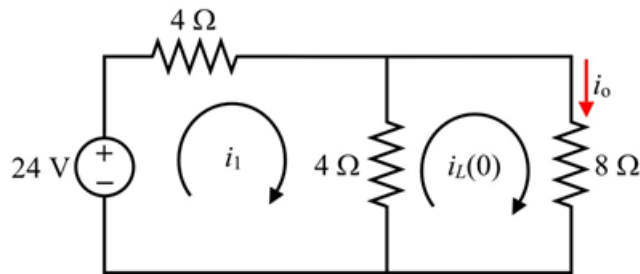


Figure 3

Step 4 of 8

Apply Kirchhoff's voltage law to left mesh.

$$4i_1 + 4(i_1 - i_L(0)) = 24$$

$$(4 + 4)i_1 - 4i_L(0) = 24$$

$$8i_1 - 4i_L(0) = 24$$

$$2i_1 = 6 + i_L(0)$$

$$i_1 = \frac{6 + i_L(0)}{2} \dots\dots (1)$$

Step 5 of 8

Apply Kirchhoff's voltage law to right mesh.

$$4(i_L - i_1) + 8i_L(0) = 0$$

$$-4i_1 + 12i_L(0) = 0$$

Substitute $\frac{6 + i_L(0)}{2}$ for i_1 in the above equation.

$$-4\left(\frac{6 + i_L(0)}{2}\right) + 12i_L(0) = 0$$

$$-12 - 2i_L(0) + 12i_L(0) = 0$$

$$10i_L(0) = 12$$

$$i_L(0) = 1.2 \text{ A}$$

Step 6 of 8

At $t = 0$, the switch in the circuit is opened. Draw the modified circuit diagram:

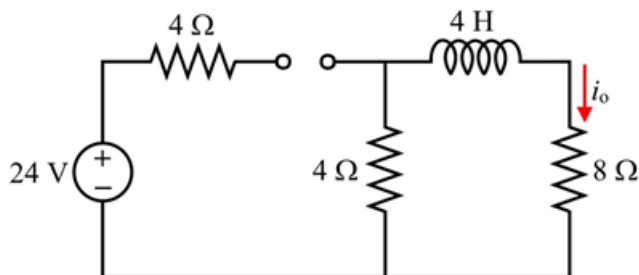


Figure 4

Step 7 of 8

For $t > 0$, the switch is open and voltage source is disconnected. Now the circuit is source free RL circuit.

Looking from the terminals of the inductor in Figure 4, $4\ \Omega$ and $8\ \Omega$ resistors are connected in series. Calculate the equivalent resistance of series resistors.

$$\begin{aligned} R_{eq} &= 4\ \Omega + 8\ \Omega \\ &= 12\ \Omega \end{aligned}$$

Calculate the time constant.

$$\begin{aligned} \tau &= \frac{L}{R_{eq}} \\ &= \frac{4\ \text{H}}{12\ \Omega} \\ &= \frac{1}{3}\ \text{s} \end{aligned}$$

Step 8 of 8

Write the expression for current through inductor for $t > 0$ of a source free circuit.

$$i_L(t) = i_L(0) e^{-\frac{t}{\tau}}$$

Calculate the current through inductor for $t > 0$.

$$\begin{aligned} i_L(t) &= (1.2) e^{-\left(\frac{t}{\frac{1}{3}}\right)} \\ &= 1.2 e^{-3t}\ \text{A} \end{aligned}$$

The current through inductor and current through $8\ \Omega$ both are same, because both are connected in series.

$$\begin{aligned} i_o(t) &= i_L(t) \\ &= 1.2 e^{-3t}\ \text{A} \end{aligned}$$

Therefore, the current, i_o for $t > 0$ is $\boxed{1.2 e^{-3t}\ \text{A}}$.